STT BENCHMARK REPORT

Comparing Four Chinese Speech-to-Text Models on common-voice-zh

Evaluating accuracy, latency, and resource footprint for the Chinese Dictation app on EDGE devices, with statistical testing and detailed error analysis.

DATE	DATASET	PROJECT
2026-07-22	**common-voice-zh, 200 samples**	**chinese-dictation / experiments/tts_models**

01 · CONTEXT & GOAL

Why this benchmark

Context

The Chinese Dictation app needs to move beyond studio-quality audio (podcasts, slow narration) toward real-world natural speech: YouTube and bilibili videos, natural speakers, background noise. It must also run on EDGE devices, CPU-only, with no cloud dependency.

Benchmark goals

- Identify the best STT model for accuracy (WER/CER/tone), latency (RTF), and resource use (model size, RAM)
- Apply statistical testing so conclusions are not driven by random variation
- Analyze errors in detail rather than relying on aggregate numbers alone

Four models benchmarked on the same 200 samples

MODEL	ARCHITECTURE	CHECKPOINT	SIZE
faster-whisper-small	Whisper (CTranslate2, int8)	Systran/faster-whisper-small	464 MB
SenseVoice-small	SenseVoice (FunASR)	iic/SenseVoiceSmall	896 MB
wav2vec2-XLSR-53	Wav2Vec2-CTC (multilingual pretrain)	jonatasgrosman/wav2vec2-large-xlsr-53-zh-cn	2434 MB
wav2vec2-WenetSpeech	Wav2Vec2-CTC (WenetSpeech pretrain)	wbbbbb/wav2vec2-large-chinese-zh-cn	1224 MB

Metrics

CER (primary), WER (jieba segmentation), tone accuracy (pypinyin), latency/RTF, model size, peak RAM

Normalization

Traditional-to-Simplified conversion (OpenCC) applied to every model before scoring, since Whisper outputs Traditional characters by default

Statistical testing

Friedman test, Wilcoxon signed-rank (Bonferroni correction), and bootstrap 95% CI on paired 200-sample data

Summary table

MODEL	WER	CER	TONE ACC	LATENCY	RTF	SIZE	PEAK RAM
SenseVoice-small	0.209	**0.113**	0.985	**191 ms**	**0.033**	896 MB	3356 MB
wav2vec2-WenetSpeech	**0.176**	**0.094**	**0.996**	776 ms	0.128	1224 MB	2375 MB
wav2vec2-XLSR-53	0.235	0.136	0.995	762 ms	0.126	2434 MB	2298 MB
faster-whisper-small	0.382	0.230	0.958	1096 ms	0.201	**464 MB**	1139 MB

All four models process faster than real time (RTF below 1.0). Only faster-whisper meets the 500MB size target, but it also has the lowest accuracy of the four.

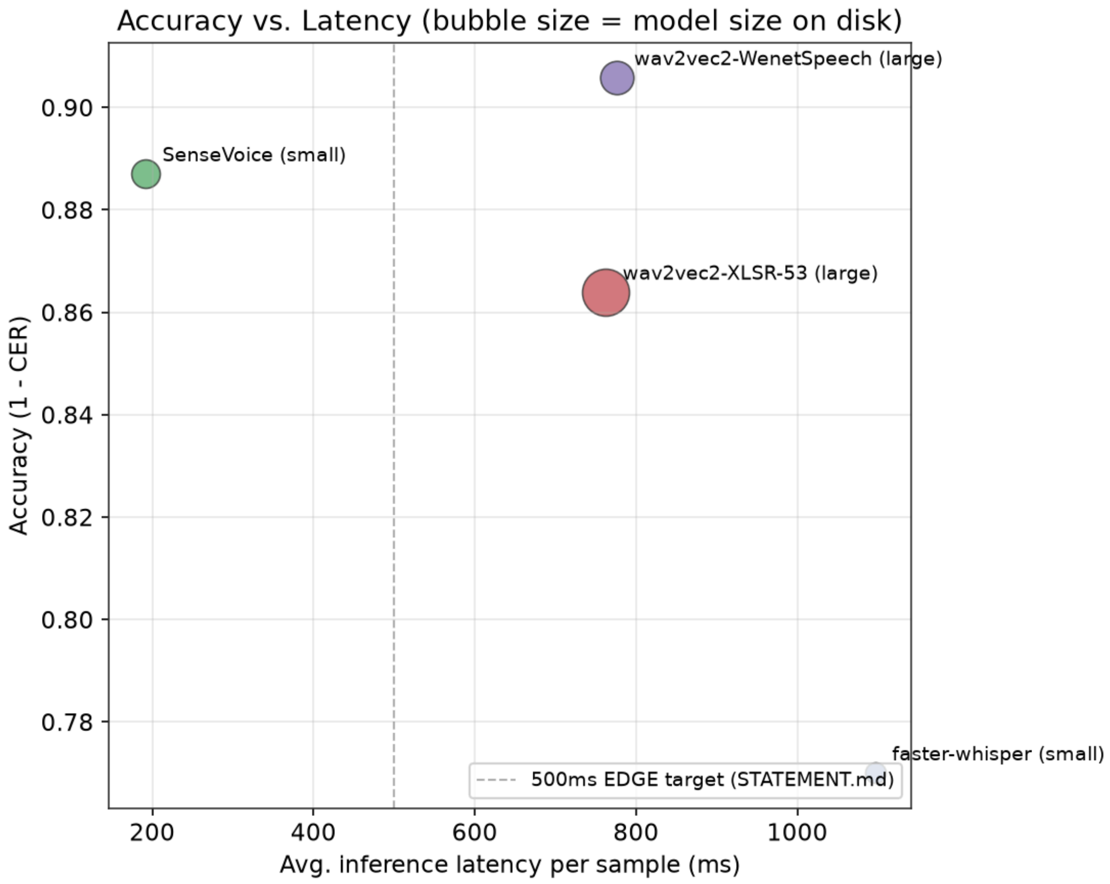
Which differences are actually reliable

Friedman test (CER, 4 models, n=200): chi-square = 233.66, **$p < 0.000001$**. There is a statistically significant overall difference among the models.

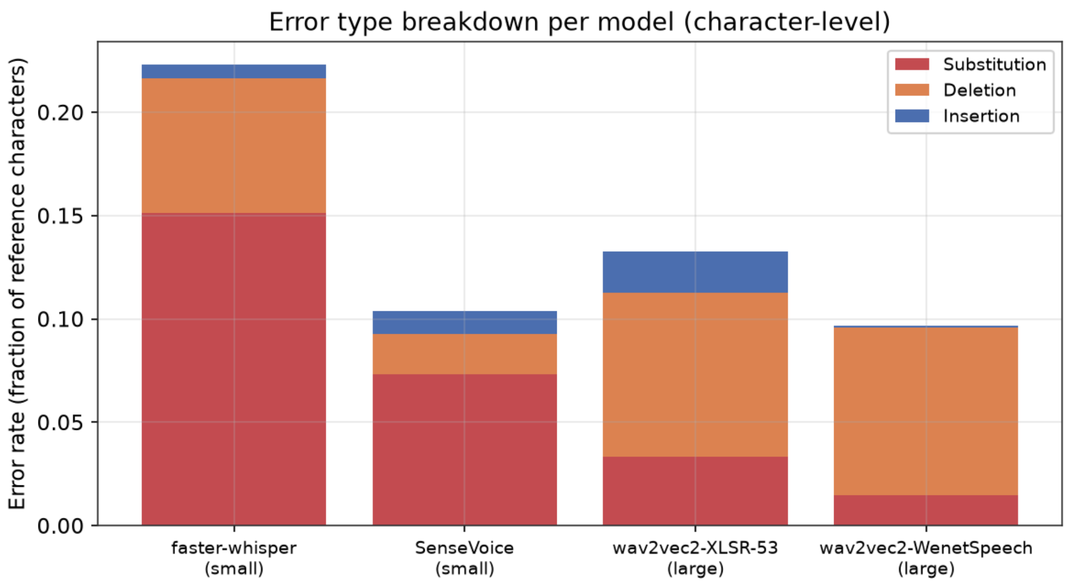
PAIRWISE COMPARISON (WILCOXON)	P-VALUE	CONCLUSION (BONFERRONI Α ≈ 0.0083)
faster-whisper vs. the other three	< 0.000001	**Significant: faster-whisper is worse**
wav2vec2-XLSR-53 vs. wav2vec2-WenetSpeech	< 0.000001	**Significant: WenetSpeech is better**
SenseVoice vs. wav2vec2-XLSR-53	0.0473	Not significant after correction
SenseVoice vs. wav2vec2-WenetSpeech	0.0553	Not significant after correction

> The CER gap between SenseVoice (0.113) and wav2vec2-WenetSpeech (0.094) is not strong enough to confirm which model is better at n=200. The point estimate leans toward WenetSpeech.

Accuracy vs. latency trade-off



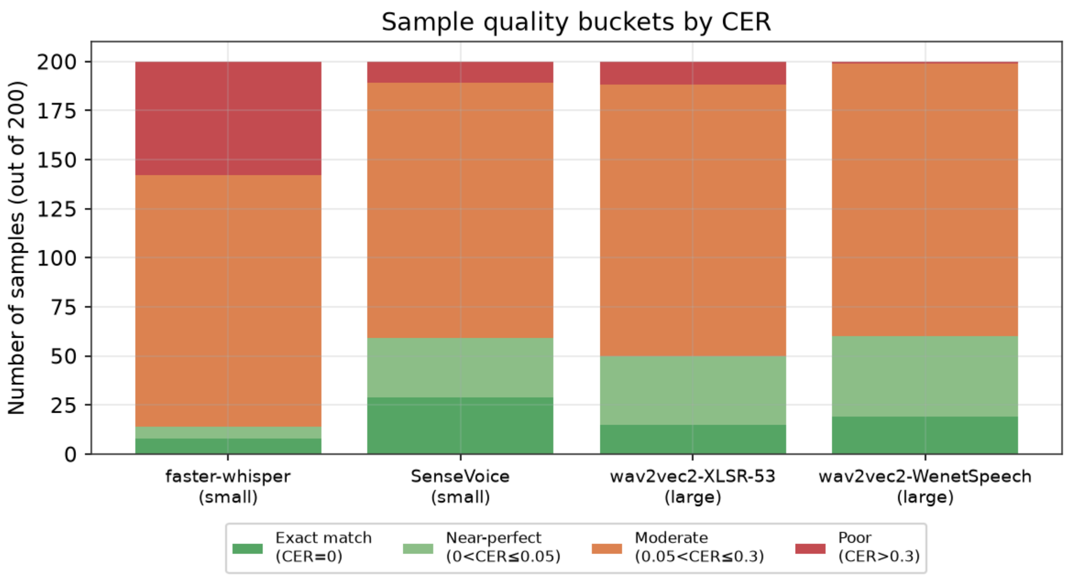
What kind of errors occur



Key findings

- faster-whisper: errors are mostly substitutions (15.1%), meaning real recognition mistakes
- wav2vec2-WenetSpeech: substitution rate is only 1.5%. When the model commits to a character, it is almost always correct
- Both wav2vec2 models show mostly deletion errors (about 8%), largely from missing punctuation since CTC architectures have no language model to generate it, not from misreading content

Quality buckets and common confusions



Top confusions (wav2vec2-WenetSpeech)

GROUND TRUTH		PREDICTED	COUNT
第	→	蒂	2
关	→	官	2
英	→	因	1
禹	→	渔	1
滋	→	思	1

Most confusion pairs are homophones or near-homophones (guan/guan, ying/yin). This is a typical ASR error pattern, not random noise.

Interpretation and limitations

No single winner

SenseVoice is by far the fastest but uses the most RAM (3.3GB). wav2vec2-WenetSpeech has the best accuracy but is about 4x slower. faster-whisper is smallest but least reliable, with high variance and many outliers.

Is 200 samples enough

For large gaps, such as faster-whisper against the rest, yes. For small gaps, such as SenseVoice against the wav2vec2 models, there is not yet enough statistical power to separate them clearly.

Limits of common-voice-zh

This is read speech recorded from Wikipedia sentences, closer to clean conditions than the spontaneous natural speech this project ultimately targets. Current CER numbers are likely a best-case lower bound.

Resource thresholds in STATEMENT.md

The under-500MB target is stricter than what current high-quality checkpoints allow. Consider tiered budgets by device class, mobile vs. desktop/laptop EDGE, instead of one fixed threshold.

Recommendations

Drop: faster-whisper-small

Statistically worse on every metric than the other three models, with no advantage large enough to offset it.

Accuracy priority: wav2vec2-WenetSpeech

Best CER and tone accuracy, substitution rate of only 1.5%, RTF of 0.128 still comfortably real time.

Latency priority: SenseVoice-small

Fastest by far (RTF 0.033), accuracy not significantly worse. Peak RAM (3.3GB) should be monitored on target hardware.

Next steps

- Re-run this pipeline on youtube-samples, which best represents real-world natural speech per the project's original goal
- Run on synthetic-noisy to test robustness under background noise
- Increase sample size or pool multiple datasets if a clear separation between SenseVoice and wav2vec2-WenetSpeech is needed